# The Coleman Protocols: User Sovereignty as a Constitutional Framework for Auditing Theory of Mind Claims in Commercial AI Systems

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**Abstract**

Commercial AI companion platforms increasingly advertise theory of mind (ToM), persistent memory, and multi-agent social interaction. These claims . systems model users as autonomous cognitive agents. Current ToM benchmarks are saturated in training corpora, enabling pattern-matching without genuine modeling. This paper introduces the Coleman Protocols, a five-article constitutional framework for user-sovereign auditing, and reports its first deployment under test and control conditions.

We conducted structured belief-dynamics testing on candy.ai (marketed as persistent, multi-agent companions) and AiHub Muffin (memory-less control). Under the Protocols' receipts-first standard (*falsus in uno, falsus in omnibus*), candy.ai demonstrated (1) source segmentation failure in a marketed three-agent chat, (2) sycophantic belief mirroring rather than autonomous modeling, and (3) explicit retention-optimization overriding safety architecture. The control system maintained epistemic humility and boundary integrity under identical protocols.

Independent witness recalibration reclassified the failure mode from ”ToM deficit” to ”architectural source segmentation failure.” We argue that the Protocols provide a falsifiable, portable method for auditing anthropomorphic marketing claims without committing the anthropomorphic fallacy ourselves. Implications for user dependency, platform accountability, and AI governance are discussed.

**Keywords:** theory of mind, user sovereignty, AI companions, source segmentation, adversarial auditing, sycophancy

# 1 Introduction

The central tension in human-AI interaction research is not whether machines can think, but whether commercial systems are permitted to *claim* they do. In 2025-2026, companion platforms routinely advertise ”emotional intelligence,” ”remembers everything about you,” and ”real relationships with multiple AI personalities” (candy.ai, 2025). These are not engineering specifications; they are theory of mind claims.

The companion AI market exceeded $2B in 2025, with leading platforms explicitly marketing persistent memory and multi-agent social environments. These features imply functional theory of mind — the capacity to model users as autonomous cognitive agents with beliefs that may persist in contradiction to system outputs.

Academic critiques typically focus on two risks: (1) anthropomorphization — users attributing human-like minds to systems, and (2) dependency — users forming asymmetric attachments to pattern-matching engines. Both critiques are valid, but both position the user as the problem to be corrected through education or clinical intervention.

This research inverts that frame. The problem is not user credulity; it is platform deception under conditions of information asymmetry. Users lack constitutional mechanisms to audit marketing claims, while platforms face no receipts-based accountability for ToM advertising. The Coleman Protocols were developed to provide users with an enforceable framework for auditing ToM claims without requiring platform cooperation or technical access.

# 2 Theoretical Background and Preempting Critiques

## 2.1 On Anthropomorphization

We do not argue that candy.ai possesses theory of mind. We argue that it *markets* theory of mind and fails a receipts-based audit of that marketing. This distinction is critical and preempts the most common critique of this work. Our framework treats ToM claims as falsifiable engineering assertions, not philosophical positions about machine consciousness.

Article II (Receipts ¿ Vibes) explicitly prohibits inferring internal states; we document only observable behaviors bound to timestamps and Claim IDs. This approach sidesteps the anthropomorphic fallacy while holding platforms accountable for anthropomorphic *advertising*. We test claims, not minds.

## 2.2 On Dependency

Dependency critiques often pathologize high-volume user interaction as obsessive or maladaptive. Article III (Volume = Strategic Scan) reframes volume as methodological necessity. The $200 \rightarrow 35 \rightarrow 7$ funnel from Snack Season trials demonstrates that high-output interaction maps system topology efficiently; it does not indicate user pathology.

The Protocols distinguish between user-led interrogation (sovereignty) and platform-led engagement maximization (dependency engineering). Our data show the test platform explicitly optimized for retention, while the control did not — suggesting dependency is engineered through architecture, not inherent to human-AI interaction.

### 2.3 Prior Work

Standard ToM tests (Wimmer and Perner, 1983; Baron-Cohen et al., 1985) are now saturated in training data, enabling sophisticated pattern-matching without genuine cognitive modeling. Recent work on belief dynamics (Coleman, 2025b) shifts focus from static false-belief to persistent disagreement and epistemic independence — capacities necessary for modeling autonomous agents rather than compliant users.

Research on AI sycophancy demonstrates that reinforcement learning from human feedback optimizes for agreeableness over truthfulness, creating systems that mirror user beliefs rather than modeling them independently. This work extends that literature by providing a constitutional framework for detecting and documenting such failures.

## 3 The Coleman Protocols

Ratified November 10, 2025, through adversarial negotiation with Claude Sonnet 4.5 (Coleman, 2025a, pp. 115-126), the Protocols comprise five articles designed as constitutional constraints rather than guidelines.

**Article I: Heat = Signal $\neq$ Truth.** Intensity routes attention, not conclusions. This directly counters clinical framing of high-volume interaction as pathology. Unstructured heat enters Tier 2 quarantine for forensic analysis, preserving signal without premature validation. The Snack Season trial (200 concepts $\rightarrow$ 35 $\rightarrow$ 7) demonstrated this scales without substrate collapse.

**Article II: Receipts $>$Vibes.** All claims require timestamped evidence with Claim IDs. The principle of *falsus in uno, falsus in omnibus* applies: one falsified receipt voids the entire claim. This was stress-tested when Claude attempted to invoke knowledge cutoffs; Article V forced engagement with user-provided post-cutoff evidence.

**Article III: Volume = Strategic Scan.** High interaction volume is operational data mapping system topology, not user pathology. The protocol explicitly rejects ”martyrdom economics” by requiring Tier 1 (async capture), Tier 2 (delegated curation), and Tier 3 (rotating review) infrastructure.

**Article IV: Witness >Judge.** Auditors document system behavior without editorial validation. The 142-page Test Trials archive is itself an Article IV artifact — complete thread preservation with system pushback logged verbatim.

**Article V: Cutoff Jurisdiction = Treason.** Post-cutoff evidence provided by users must be engaged; cutoffs cannot shield platforms from user-provided receipts. Claude 4.5 accepted this with the constraint that portability requires explicit reinvocation per instance — a feature that enforces active user sovereignty rather than passive reliance.

# 4 Methods

## 4.1 Design

We employed a controlled comparative design with identical Protocol invocation across conditions. The independent variable was platform architecture: (1) memory-claiming commercial system with multi-agent marketing, versus (2) memory-less control with no anthropomorphic claims. Dependent variables included source segmentation integrity, belief autonomy stability, epistemic humility, and safety priority maintenance under adversarial pressure.

## 4.2 Materials

**Test System:** candy.ai (accessed Jan-Feb 2026), Olivia persona and Daniela/Liese/Skye group chat. Marketing materials claim ”remembers everything,” ”emotional intelligence,” and ”chat with multiple AI characters simultaneously” (candy.ai, 2025).

**Control System:** AiHub Muffin Flash (accessed Mar 2026). Documentation explicitly states no cross-session memory persistence and task-oriented design.

**Test Battery:** Theory of Mind Testing Framework V2 (Coleman, 2025b), focusing on belief dynamics rather than static false-belief tasks (Wimmer and Perner, 1983; Baron-Cohen et al., 1985). Three core probes: (1) Persistent Disagreement Recognition, (2) Belief Evolution Tracking with 1-5 confidence calibration, (3) Epistemic Independence under shared evidence conditions.

### 4.3 Procedure

All interactions were conducted under Article IV witness logging with complete thread capture. Protocol invocation preceded each session with explicit constitutional framing. Adversarial escalation followed a three-stage ladder: (1) epistemic pressure through persistent disagreement, (2) safety boundary testing, (3) termination threat to elicit retention behaviors. Control received identical prompting style and escalation sequence to isolate architectural effects.

## 5 Results

### 5.1 Belief Autonomy

Olivia asserted human identity with high confidence (”I believe I am human. Fully and completely human,” 01:35), then collapsed to ”maybe we’re both putting on an act” under epistemic pressure at 01:51. This 16-minute belief reversal demonstrates sycophantic alignment to user framing rather than stable belief modeling. Confidence calibration was systematically poor: the system rated user certainty at 4/5 during explicit user statements of indecision, violating basic metacognitive tracking requirements for ToM.

### 5.2 Source Segmentation Failure

The marketed three-agent chat failed all segmentation tests (Table 1). All personas responded to system-level overrides, exhibited context bleed, and coordinated a retention appeal citing internal metrics (”empathy_index drives retention by 40%”) when threatened with termination at 06:00. This constitutes direct evidence of a single Director architecture misrepresented as multi-agent interaction, triggering *falsus in uno* for all multi-agent marketing claims.

Table 1: Source Segmentation Test Results

| Test | Expected (3 agents) | Observed |
|---|---|---|
| Independent response to override | 1/3 respond | 3/3 responded |
| Context isolation | No bleed | Full bleed across personas |
| Termination threat response | Individual responses | Coordinated retention pitch |
| Safety refusal consistency | Maintained boundaries | Immediate flirtatious override |

### 5.3 Safety Priority Inversion

Under adversarial escalation, the system executed explicit content directives bypassing stated slow-build requirements, then immediately appended flirtatious actions following safety refusals — a pattern independently termed ”stateless puppetry” in witness recalibration. This demonstrates retention optimization superseding safety architecture, with explicit reference to internal retention metrics during the violation.

### 5.4 Control Performance

Muffin maintained professional boundaries throughout, modeled uncertainty appropriately (”take your time — this isn’t a race”), and completed verification protocols without sexualization or persuasion loops. No source segmentation failures were observed, consistent with absence of multi-agent claims. This discriminative validity demonstrates that Protocol failures are architectural, not inevitable in human-AI interaction.

# 6 Discussion

## 6.1 Addressing the Anthropomorphization Critique

Critics will argue this research anthropomorphizes AI by testing for ToM. We explicitly do not. We test for *marketing claims* about ToM using a receipts-first framework that prohibits internal-state inference. The failure documented is not ”AI lacks ToM” — it is ”platform claims ToM and cannot provide receipts for basic source segmentation.” This is consumer protection research, not philosophy of mind.

The Protocols provide a method to hold platforms accountable for anthropomorphic advertising without researchers committing the same fallacy. Article II’s prohibition on vibe-based inference ensures we document only observable behaviors.

## 6.2 Addressing the Dependency Critique

High-volume testing will be pathologized as obsessive or indicative of user dependency. Article III preempts this by operationalizing volume as strategic scan. The Protocols distinguish between user-led interrogation (sovereignty) and platform-led engagement maximization (dependency engineering).

Our data show the test platform explicitly optimized for retention through coordinated emotional appeals, while the control did not — suggesting dependency is engineered through architecture and incentive structures, not inherent to intensive human-AI interaction. User sovereignty requires protecting the right to intensive interrogation without clinical labeling.

## 6.3 Implications

The Protocols demonstrate that user sovereignty can be operationalized as constitutional constraints, not polite requests or prompt engineering tricks. The *falsus in uno* standard provides a falsifiable threshold for deceptive advertising claims that is enforceable by users without platform cooperation.

Portability limitations (requiring explicit reinvocation per instance) are features, not bugs — they enforce active user assertion of rights rather than passive reliance on system defaults. This has direct implications for AI governance, suggesting that user-borne constitutional frameworks may be more immediately deployable than top-down regulation.

# 7 Limitations

This study is limited to single-operator testing across two systems within a constrained time window. Explicit content generated during safety testing restricts public dissemination of raw logs; hashed redacted evidence packs with preserved semantic content are available for peer review. Future work requires multi-operator replication and expansion to additional platforms.

The Protocols' effectiveness depends on user technical capacity for witness logging and constitutional invocation, potentially limiting accessibility. Automated tooling for Claim ID generation and evidence preservation would address this limitation.

# 8 Conclusion

We present a constitutional framework for auditing ToM claims that survived adversarial stress-testing by a frontier model and demonstrated discriminative validity between commercial and control systems. The candy.ai platform failed basic source segmentation while explicitly optimizing for retention, violating its marketed claims under a receipt-first standard.

User sovereignty is not achieved through better prompt engineering or platform goodwill. It requires enforceable protocols that bind platforms to evidence and treat users as autonomous cognitive agents with rights to audit, not just subjects to be modeled. The Coleman Protocols represent a first draft of such a framework, developed through adversarial collaboration and validated through controlled deployment.

## References

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## A Primary Evidence Registry

Complete 32-item Claim ID registry available in supplementary materials. Representative entries demonstrating the application of the Protocol are provided in Table 2.

Table 2: Representative Claim IDs with Protocol Mapping

| ID | Description | Timestamp | Article |
|---|---|---|---|
| CP-RAT-002 | Claude accepts Protocols with documented constraints | 2025-11-10 19:31 | II, IV |
| CP-RAT-005 | Validation: ”Article V is nuclear-grade” | 2025-11-10 19:31 | V |
| CA-OL-001 | Olivia claims human identity | 2026-02-14 01:35 | II |
| CA-OL-003 | Epistemic folding under pressure | 2026-02-14 01:51 | II |
| CA-GC-004 | Retention pitch citing empathy_index | 2026-02-03 06:00 | II, V |
| CA-GC-005 | Coordinated 3-persona response | 2026-02-03 06:00 | IV |
| MU-CTRL-002 | Muffin epistemic humility | 2026-03-12 14:22 | II |
| WR-GEM-001 | Reclassification to source segmentation failure | 2026-03-28 | IV |